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Course: Calculus Beta

Assignment: MyLab Math - Uge 12

Find all the second-order partial derivatives of the function.

$$r(x,y) = \ln(6x + 7y)$$

A second-order derivative is produced by differentiating a function twice. The function r has four second-order derivatives $\frac{\partial^2 r}{\partial x^2}$, $\frac{\partial^2 r}{\partial y \partial x}$, $\frac{\partial^2 r}{\partial x \partial y}$, and $\frac{\partial^2 r}{\partial y^2}$. Evaluate each of these second-order derivatives in order.

Start with the first second-order derivative $\frac{\partial^2 r}{\partial x^2}$. It is defined as $\frac{\partial^2 r}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial r}{\partial x} \right)$.

First differentiate r with respect to x . Treat y as a constant and differentiate the given function r with respect to x in the usual way.

$$\begin{aligned} \frac{\partial r}{\partial x} &= \frac{\partial}{\partial x} \ln(6x + 7y) \\ &= \frac{1}{6x + 7y} \frac{\partial}{\partial x} (6x + 7y) \end{aligned}$$

Then evaluate the partial derivative in the result.

$$\begin{aligned} \frac{\partial r}{\partial x} &= \frac{1}{6x + 7y} \frac{\partial}{\partial x} (6x + 7y) \\ &= \frac{1}{6x + 7y} (6) \\ &= \frac{6}{6x + 7y} \end{aligned}$$

Next differentiate the result with respect to x to find the second-order derivative.

$$\begin{aligned} \frac{\partial^2 r}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{6}{6x + 7y} \right) \\ &= - \frac{36}{(6x + 7y)^2} \end{aligned}$$

Now find the second second-order derivative $\frac{\partial^2 r}{\partial y \partial x}$. It is defined as $\frac{\partial^2 r}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial r}{\partial x} \right)$.

Differentiate the partial derivative $\frac{\partial r}{\partial x} = \frac{6}{6x + 7y}$ with respect to y .

$$\begin{aligned} \frac{\partial^2 r}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(\frac{6}{6x + 7y} \right) \\ &= - \frac{42}{(6x + 7y)^2} \end{aligned}$$

Next find the third second-order derivative $\frac{\partial^2 r}{\partial x \partial y}$. It is defined as $\frac{\partial^2 r}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial r}{\partial y} \right)$.

Use the mixed derivative theorem. The mixed derivative theorem states that if $f(x,y)$ and its partial derivatives f_x , f_y , f_{xy} , and f_{yx} are defined throughout an open region containing a point (a,b) and are all continuous at (a,b) , then $f_{xy}(a,b) = f_{yx}(a,b)$.

The partial derivatives $\frac{\partial r}{\partial x}$, $\frac{\partial r}{\partial y}$, $\frac{\partial^2 r}{\partial y \partial x}$, and $\frac{\partial^2 r}{\partial x \partial y}$ are defined and continuous for all values for which the given function r is defined, that is,

where $6x + 7y > 0$. Use the partial derivative $\frac{\partial^2 r}{\partial y \partial x} = - \frac{42}{(6x + 7y)^2}$ to find the other mixed derivative in this domain.

$$\frac{\partial^2 r}{\partial x \partial y} = -\frac{42}{(6x + 7y)^2}$$

Finally, find the fourth second-order derivative $\frac{\partial^2 r}{\partial y^2}$. It is defined as $\frac{\partial^2 r}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial r}{\partial y} \right)$.

First differentiate r with respect to y . Treat x as a constant and differentiate the given function r with respect to y in the usual way.

$$\begin{aligned} \frac{\partial r}{\partial y} &= \frac{\partial}{\partial y} \ln(6x + 7y) \\ &= \frac{1}{6x + 7y} \frac{\partial}{\partial y}(6x + 7y) \end{aligned}$$

Then evaluate the partial derivative in the result.

$$\begin{aligned} \frac{\partial r}{\partial y} &= \frac{1}{6x + 7y} \frac{\partial}{\partial y}(6x + 7y) \\ &= \frac{1}{6x + 7y} (7) \\ &= \frac{7}{6x + 7y} \end{aligned}$$

Next differentiate the result with respect to y to find the second-order derivative.

$$\begin{aligned} \frac{\partial^2 r}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{7}{6x + 7y} \right) \\ &= -\frac{49}{(6x + 7y)^2} \end{aligned}$$

Thus, the four second-order derivatives are shown below.

$$\begin{aligned} \frac{\partial^2 r}{\partial x^2} &= -\frac{36}{(6x + 7y)^2} \\ \frac{\partial^2 r}{\partial y^2} &= -\frac{49}{(6x + 7y)^2} \\ \frac{\partial^2 r}{\partial y \partial x} &= \frac{\partial^2 r}{\partial x \partial y} = -\frac{42}{(6x + 7y)^2} \end{aligned}$$